

DICKSON'S CONJECTURE AND INFINITE MULTIDIMENSIONAL SUM-SETS WITHIN THE PRIMES

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ABSTRACT. Assuming Dickson's conjecture, we show that for every prime p and every $n \leq p - 2$ the set P of primes contains a sum-set $\{p\} + E_0 + \cdots + E_n$ with each E_i an infinite set of even numbers containing 0. Unconditionally, no such configuration exists once $n \geq p - 1$, so the result is sharp. In particular the primes contain no infinite-dimensional sum-set, although random models capturing the density behavior of the primes do; in this respect the primes fail to behave like a typical prime-like random set.

1. INTRODUCTION

We shall show that Dickson's conjecture concerning the set P of prime numbers implies that P contains certain strikingly highly structured subsets. Precise statements follow once we establish terminology and notation. Throughout, $\mathbb{N}$ denotes the natural numbers including 0, and $2\mathbb{N}$ the even natural numbers.

Given a family $\{A_i\}$ of subsets of $\mathbb{N}$, write $\bigoplus A_i$ for the subset of $\prod A_i$ consisting of vectors with all but finitely many coordinates equal to 0. (For infinite families $\{A_i\}$ we always assume all but finitely many sets contain 0.) The *sum-set* of $\{A_i\}$, written $\sum A_i$ or $A_0 + A_1 + \cdots$, means the set $\{\sum_i x_i \mid (x_i) \in \bigoplus A_i\}$.

The language of sum-sets helps express notions of additive number theory. For example, the famous Twin Prime Conjecture predicts an infinite subset $T \subset P$ such that $T + \{0, 2\} \subset P$. A long record of intractability has not prevented the Twin Prime Conjecture from spawning many (of course conjectural) generalizations, perhaps most famously Schinzel's Hypothesis H [5]. Here we develop a formal consequence of an older generalization of intermediate ambition, namely Dickson's conjecture [2]. Dickson's conjecture concerns finite families $\{a_i m + b_i\}$ of linear polynomials in m with $a_i \geq 1$ and $b_i \in \mathbb{N}$, having the property that no one prime divides $\prod_i (a_i n + b_i)$ for every $n \in \mathbb{N}$; the conjecture predicts infinitely many values of m making all the quantities $a_i m + b_i$ simultaneously prime.

Now we can state our results.

Theorem 1. *Assume Dickson's conjecture. Then for every prime p and every integer n with $1 \leq n \leq p - 2$ there exist infinite sets $E_0, \dots, E_n \subset 2\mathbb{N}$, each containing 0, such that*

$$\{p\} + \sum_{i=0}^n E_i \subset P.$$

The restriction $n \leq p - 2$ cannot be relaxed, and in particular no infinite-dimensional analogue can hold:

Proposition 2. *Let p be prime and let $E_0, \dots, E_{p-1} \subset \mathbb{N}$ each contain 0 and at least one positive element. Then $\{p\} + \sum_{i=0}^{p-1} E_i \not\subset P$. Consequently, for no prime p and no family*

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$\{E_i\}_{i \in I}$ with $|I| \geq p$ (in particular, no infinite family) of sets each containing 0 and a positive element can one have $\{p\} + \sum_{i \in I} E_i \subset P$.

Together the two statements give a sharp dichotomy: under Dickson's conjecture, a sum-set of the form $\{p\} + E_0 + \cdots + E_n$ (with $0 \in E_i$, E_i infinite) lies within the primes exactly when $n \leq p - 2$.

We place these statements in context. The existence assertion of Theorem 1, viewed simply as an $(n + 1)$ -fold conditional sum-set within the primes, is not new: Granville [3] showed already in 1990, under the Dickson–Hardy–Littlewood conjecture, that the primes contain k -fold sum-sets $A_1 + \cdots + A_k$ of infinite sets for every k . Indeed the unanchored form of Theorem 1 falls out of Granville's theorem in a few lines: with $A_0 + \cdots + A_n \subseteq P$ as [3] provides, choose $a_i \in A_i$ and put $p' := a_0 + \cdots + a_n$, itself a full sum and hence prime; the translates $E_i := \{b - a_i : b \in A_i, b \geq a_i, b \equiv a_i \pmod{2}\}$ are infinite, consist of even numbers, contain 0, and satisfy $\{p'\} + E_0 + \cdots + E_n \subseteq A_0 + \cdots + A_n \subseteq P$. What this derivation cannot control is the anchor: p' falls where it falls. What we believe is new here is therefore the anchored, quantitative form of the question and its answer: the exact trade-off between the anchor prime p and the achievable dimension, with the threshold $n = p - 2$ sharp in both directions. Tao and Ziegler [6] proved unconditionally that there exist infinite sets A, B with $a + b$ prime whenever $a \in A$, $b \in B$ and $a < b$; whether the full sum-set $A + B$ of two infinite sets lies in the primes remains open unconditionally. Kra, Moreira, Richter and Robertson [4] prove, conditionally on Dickson–Hardy–Littlewood, the stronger single-set statement that $P - 1$ contains all sums of at most k distinct elements of a single infinite set, for each fixed k ; they also show unconditionally that no shift of P contains an IP-set, a statement in the spirit of Proposition 2. We return in the closing remarks to a second point of interest, the contrast with random models of the primes.

2. PROOF OF PROPOSITION 2

Choose a positive element $x_i \in E_i$ for $i = 0, \dots, p - 1$ and form the prefix sums $\sigma_k := x_0 + \cdots + x_{k-1}$ for $k = 0, \dots, p$ (so $\sigma_0 = 0$). The $p + 1$ values $\sigma_k \pmod{p}$ cannot all be distinct, so $\sigma_a \equiv \sigma_b \pmod{p}$ for some $a < b$, whence

$$x_a + x_{a+1} + \cdots + x_{b-1} \equiv 0 \pmod{p}.$$

Taking the coordinates $a, \dots, b - 1$ equal to these x_i and all other coordinates equal to 0 produces an element of $\{p\} + \sum_i E_i$ congruent to 0 modulo p and strictly larger than p ; that element is composite. The final assertion follows by restricting attention to any p of the sets. $\square$

Remarks. The pigeonhole step is the statement that the Davenport constant of $\mathbb{Z}/p\mathbb{Z}$ equals p ; the bound is attained (take every $x_i \equiv 1 \pmod{p}$ with only $p - 1$ sets), which is why the threshold sits at $p - 1$ sets rather than p . Note that the proposition is unconditional and requires neither infinitude nor evenness of the E_i .

3. PROOF OF THEOREM 1

Fix $n \geq 1$ and a prime $p > n + 1$. We must produce $n + 1$ infinite sets $E_i \subset 2\mathbb{N}$ of even numbers, each containing 0.

We notate the elements of E_i from small to large as $e_{i,j}$, $j = 0, 1, \dots$, starting with $e_{i,0} = 0$ for all i , and we determine the further elements inductively in the order

$$e_{0,1}, e_{1,1}, \dots, e_{n,1}, e_{0,2}, \dots, e_{n,2}, e_{0,3}, \dots$$

Write $E_{i,j} := \{e_{i,0}, \dots, e_{i,j}\}$ for the provisional $(j+1)$ -term initial segment of E_i . At every stage the *current sum-set* S means $\{p\} + \sum_i E_{i,j_i}$ formed from the current initial segments; the construction will maintain $S \subset P$ throughout, and $E_i := \bigcup_j E_{i,j}$ then completes the proof.

Roughly, the strategy for determining each $e_{i,j}$ comes down to applying Dickson's conjecture to translate the set

$$C^* := \{p\} + \sum_{i' < i} E_{i',j} + \sum_{i' > i} E_{i',j-1}$$

into P by some even number, which we then declare to be $e_{i,j}$: the sums newly created by the insertion are exactly $\{e_{i,j}\} + C^*$, since a sum using an older element of coordinate i was already present. The point requiring care is that we must also avoid spoiling the hypothesis of Dickson's conjecture at every *later* stage of the construction. We now set up the bookkeeping that accomplishes this.

Critical values and the residues s_q . Call an element of the current sum-set S a *critical value* if it admits a representation $p + \sum_{i \in T} x_i$ with $T \subsetneq \{0, \dots, n\}$ and $x_i \in E_{i,j_i} \setminus \{0\}$ — that is, a representation with at least one coordinate equal to 0. (If one visualizes the growing sum-set as a box with p at the origin, the critical values sit on the bounding hyperplanes through the origin.) Critical values are primes, being elements of S ; we call them *critical primes*. Once critical, always critical. The prime p itself (the empty sum) is critical from the outset. Observe that the set C^* above consists entirely of critical values: its representations omit coordinate i .

When a prime q first occurs as a critical value — its *birth* — let C_q denote the set of critical values present at that moment (including q itself) and put $N_q := |C_q|$. Provided

$$(1) \quad q > (n+1)N_q,$$

we may choose a residue $s_q \in \{1, \dots, q-1\}$ such that

$$(2) \quad (k+1)s_q \not\equiv -c \pmod{q} \quad \text{for all } 0 \leq k \leq n \text{ and all } c \in C_q :$$

each pair (k, c) forbids exactly one residue, since $k+1 \leq n+1 < p \leq q$ is invertible modulo q , and there are at most $(n+1)N_q < q$ such pairs. (Taking $k=0$ and $c=q$ shows (2) forces $s_q \not\equiv 0$, so the choice $s_q \in \{1, \dots, q-1\}$ involves no further constraint.) We treat p as born at the start, with $C_p = \{p\}$ and $N_p = 1$; here the forbidden set is exactly $\{0\}$, so *any* nonzero residue s_p works — this is precisely where the hypothesis $p > n+1$ enters.

Rule for critical primes: every element $e_{i,j}$ inserted after the birth of q satisfies $e_{i,j} \equiv s_q \pmod{q}$.

The utility of (2) rests on the following confinement lemma, which controls all critical values forever after.

Lemma 3. *Let q be a critical prime. Every critical value c' arising at any time satisfies*

$$c' \equiv c + ks_q \pmod{q} \quad \text{for some } c \in C_q \text{ and some } 0 \leq k \leq n.$$

Proof. Write $c' = p + \sum_{i \in T} x_i$ with $T \subsetneq \{0, \dots, n\}$ and each x_i positive, and let k be the number of $i \in T$ whose element x_i was inserted after the birth of q . Each such $x_i \equiv s_q \pmod{q}$ by the rule above, and $k \leq |T| \leq n$. Deleting those coordinates leaves the value $p + \sum x_i$ over the remaining (pre-birth) coordinates; all of its constituents existed at the birth of q , so it belonged to the sum-set then, and it has an empty coordinate (any element

of $\{0, \dots, n\} \setminus T$, and the deleted coordinates besides), so it is a critical value present at birth: an element of C_q . $\square$

Frozen primes. Call a prime q *urgent* at a given step if q is not critical and q does not exceed the cardinality of the upcoming sum-set. Urgency is permanent, since the sum-set only grows.

Rule for urgent primes: every element inserted from that step on satisfies $q \mid e_{i,j}$.

Note that $q = 2$ is urgent from the very first insertion (the upcoming sum-set already has two elements, and 2 is never a critical value, critical values being odd primes), so all the $e_{i,j}$ are automatically even and $E_i \subset 2\mathbb{N}$, as demanded.

The inductive step. Suppose the construction has run without incident and we are to insert $e_{i,j}$, with C^* as above. Let A be the product of all currently urgent primes and all current critical primes, and use the Chinese Remainder Theorem to find B with $B \equiv 0 \pmod{q}$ for each urgent q and $B \equiv s_q \pmod{q}$ for each critical q (the moduli are distinct primes). We seek $e_{i,j}$ of the form $Am + B$; imposing the congruence rules is then automatic. Consider the family of linear forms

$$\{Am + B + c : c \in C^*\}.$$

To apply Dickson's conjecture we must check that no prime q divides $\prod_c (Am + B + c)$ for every m .

If $q \nmid A$, then $Am + B$ runs over all residues modulo q , so it suffices that the c 's not exhaust the residue classes modulo q . But $q \nmid A$ means q is neither urgent nor critical, whence q exceeds the cardinality of the upcoming sum-set and in particular $|C^*| < q$.

If q is urgent, then $Am + B + c \equiv c \pmod{q}$, and each $c \in C^*$ is a prime different from q (the elements of C^* are critical values and q is not critical), so $q \nmid \prod_c c$.

If q is critical, then $Am + B + c \equiv s_q + c \pmod{q}$. By Lemma 3 each $c \in C^*$ satisfies $c \equiv c_0 + ks_q$ with $c_0 \in C_q$ and $0 \leq k \leq n$, so $s_q + c \equiv (k+1)s_q + c_0 \not\equiv 0 \pmod{q}$ by (2).

Dickson's conjecture therefore supplies infinitely many m making all $Am + B + c$ simultaneously prime. Among these we choose $e_{i,j} := Am + B$ so large that

- (i) every new sum $e_{i,j} + c$ exceeds every element of the old sum-set and every currently urgent or critical prime; and
- (ii) every new sum exceeds $(n+1)N + 1$, where N is the number of critical values after the insertion.

Condition (i) guarantees that urgent primes never become critical (a prime already tracked can never reappear as a new sum) and that the newly created sums are genuinely new. Condition (ii) is the *growing-room* requirement: any prime q born critical at this insertion is itself one of the new sums, so $q > (n+1)N \geq (n+1)N_q$, which is exactly (1); hence s_q exists and the bookkeeping can continue. (Several critical primes may be born at one insertion; they are then present in one another's sets C_q , as Lemma 3 requires.)

All new sums $\{e_{i,j}\} + C^*$ are prime, so the enlarged sum-set again lies within P , closing the induction. Since every coordinate receives infinitely many insertions, each $E_i = \bigcup_j E_{i,j}$ is infinite, and $\{p\} + \sum_i E_i = \bigcup S \subset P$. $\square$

4. REMARKS

Both principal results of this paper have been formally verified in Lean 4 against the Mathlib library by Aristotle (Harmonic), in collaboration with Claude (Anthropic). Proposition 2 and its finite-family consequence are verified unconditionally. Theorem 1 is verified

in conditional form: Dickson's conjecture is encoded as an explicit hypothesis of the formal statement rather than as an axiom, so the axiom footprint of the entire development consists only of Mathlib's standard foundational principles, and the unconditional results cannot silently depend on the conjecture. The formalization of the construction proceeds through a symmetric strengthening: an injective sequence of even numbers all of whose anchored sums of at most $n + 1$ distinct terms are prime, in the spirit of the $\text{FS}_{\leq k}$ -statement of [4], Appendix B.

The Bateman–Horn conjecture [1] includes a quantitative version of Dickson's conjecture, but asymptotic densities without error estimates do not help us bound the $e_{i,j}$. Lacking even conjectural tools to bound growth rates, the construction above aims for clarity rather than parsimony.

Granville's conclusion from Dickson's conjecture is dramatic: the primes contain k -fold sum-sets of infinite sets for every k . One might well ask whether so strong a conclusion ought to count as evidence *against* the conjecture. The prospect has a special appeal for the skeptic: a contradiction derived from Granville's conclusion alone would demolish Dickson's conjecture wholesale, without the labor of singling out a particular fatal admissible family. Alas for the skeptic, the probabilistic heuristic supports Granville's conclusion. Dickson's conjecture holds in almost every random model that captures the density behavior of the primes, and such random sets contain the analogous sum-sets. But curiously, the same heuristic supports a conclusion that is provably false. A typical random set with prime-like densities contains *infinite*-dimensional sum-sets: generically one may insert infinitely many dimension-increasing steps into the construction, the divisibility obstruction underlying Proposition 2 having no analogue for a random set. For the primes themselves, Proposition 2 caps the dimension of a sum-set anchored at p , unconditionally, at $p - 2$. The heuristic thus cannot distinguish the true conclusion from the false one; what separates them is arithmetic invisible to density models, namely the congruence obstruction of Proposition 2. The dichotomy of this paper locates the exact boundary at which the primes part company with their random models — and, granting Dickson, the cap $p - 2$ is attained.

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